LiFeTECH

Energy Storage Solutions



Innovative Lithium Battery System

The lithium battery system consists of rack, battery modules, battery management system (BMS), display control system and protection system. 2 level BMS design, hierarchical linkage and multiple monitoring of system status. Relay, fuse, circuit breaker and BMS constitute a comprehensive protection system integrating electrical safety and functional safety.



Configuration

Battery System

The system mainly consists of safe, efficient and long-life lithium iron phosphate cells, which are connected in series to form battery modules, and multiple modules are connected in series to form battery clusters.

Battery management system

The core components of the system can effectively protect the battery from overcharge, overdischarge and ove current. At the same time, the balanced management of the cells can ensure the safe, reliable and efficient operation of the whole system.

Power Management System

System operation data monitoring, operation strategy management, historical data record, system status record, etc.

Key product features and benefits

Safe and reliable

- High quality iron phosphate lithium battery
- Intelligent air cooling design, long service life life, stable operation
- Three level BMS design of module, cabinet and system, multiple state monitoring, hierarchical linkage, comprehensive guarantee of battery system safety.
- Battery module is designed with PC bracket and reinforce steel structure to guarantee the the highest safety of the system, in transportation, installation and operation.

Efficient and Convenient

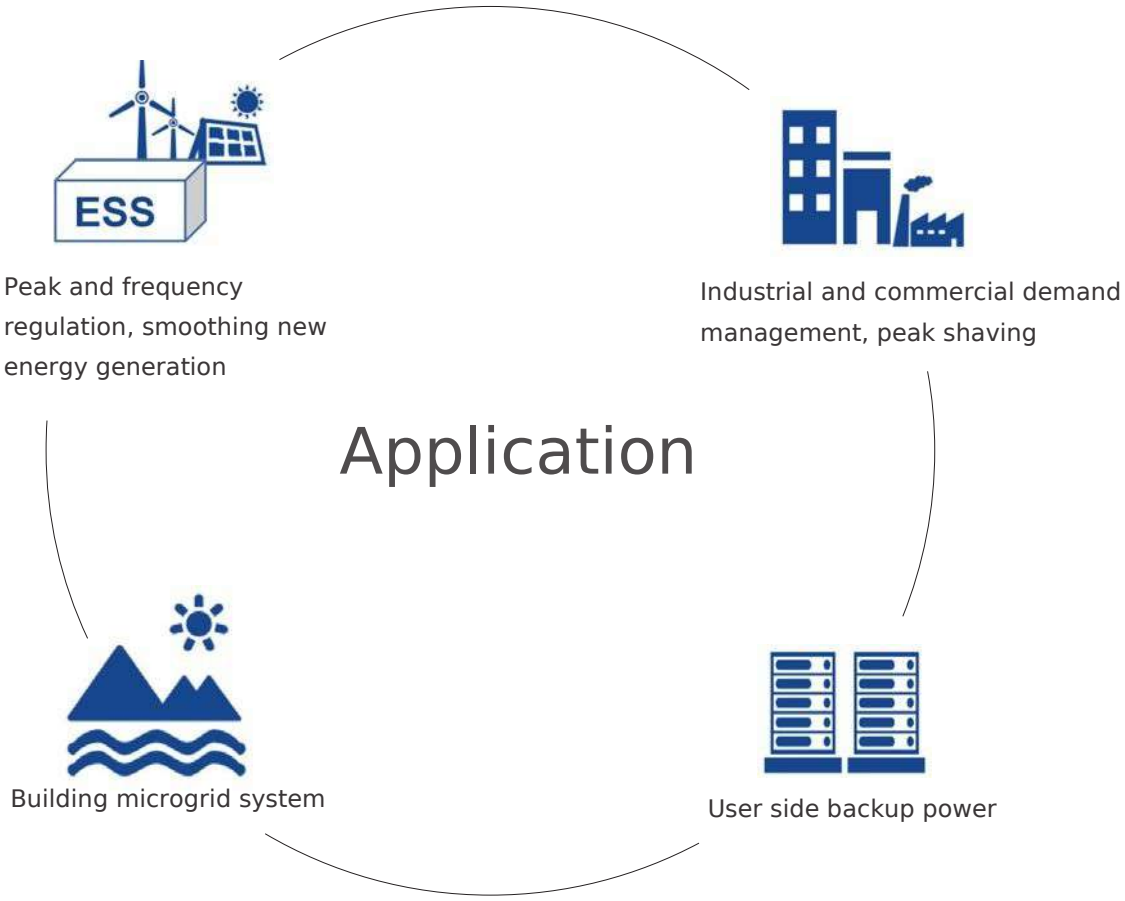
- Energy system, high energy density, high integration
- Outstanding high rate performance, maximum 2C charging and 6C discharging
- Modular design, convenient for maintenance, management and expansion.

Active equalization

- Three level BMS design, energy transferring active equalization, to overcome the impact of single cell capacity on system capacity
- The equalization accuracy is less than 2%, and the equalization capacity can reaches 10% of the rated output.

Cost optimization

- Small size, light weight, less space and installation cost
- Long cycle life, low failure rate, reduce operation and maintenance investment.



High C-rate system



EBC480/40
Battery capacity **40Ah**
Rated voltage **480Vdc**
Dimension **600*550*2000**(W*D*H)mm

EBC480/80
Battery capacity **80Ah**
Rated voltage **480Vdc**
Dimension **600*750*2000**(W*D*H)mm

EBC480/120
Battery capacity **120Ah**
Rated voltage **480Vdc**
Dimension **600*950*2000**(W*D*H)mm

EBC480/160
Battery capacity **160Ah**
Rated voltage **480Vdc**
Dimension **600*1100*2000**(W*D*H)mm

EBC768/40
Battery capacity **40Ah**
Rated voltage **768Vdc**
Dimension **1050*550*1700** (W*D*H)mm

EBC768/80
Battery capacity **80Ah**
Rated voltage **768Vdc**
Dimension **1050*750*1700** (W*D*H)mm

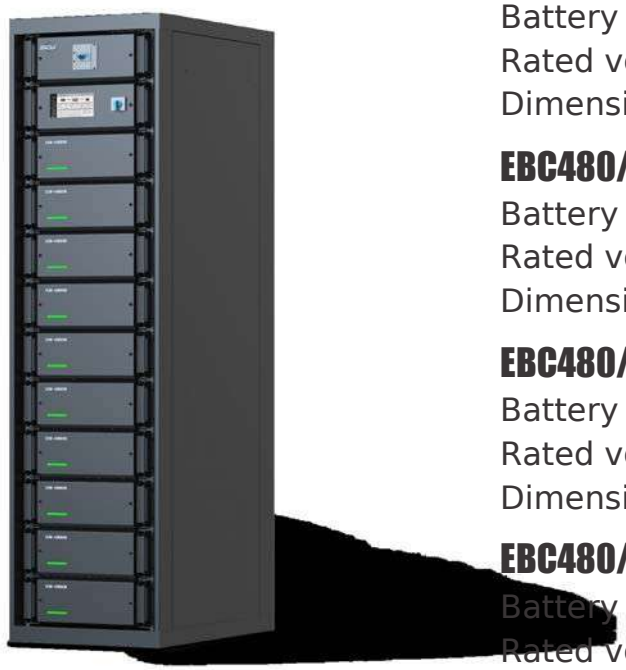
EBC768/120
Battery capacity **120Ah**
Rated voltage **768Vdc**
Dimension **1050*950*1700** (W*D*H) mm

EBC768/160
Battery capacity **160Ah**
Rated voltage **768Vdc**
Dimension **1050*1100*1700** (W*D*H) mm



Model	EBC480/40	EBC480/80	EBC480/120	EBC480/160	EBC768/40	EBC768/80	EBC768/120	EBC768/160
Battery capacity	40Ah	80Ah	120 Ah	160 Ah	40Ah	80Ah	120Ah	160Ah
Rated voltage	480V	480V	480V	480V	768V	768V	768V	768V
Rated capacity	19.2kWh	38.4kWh	57.6kWh	76.8kWh	30.72kWh	61.44kWh	92.16kWh	122.88kWh
Battery voltage range(V)	420 ~ 547.5	420 ~ 547.5	420 ~ 547.5	420 ~ 547.5	672 ~ 876	672 ~ 876	672 ~ 876	672 ~ 876
Max. Continuous charging current	80A (2C)	160A (2C)	240A (2C)	320A (2C)	80A (2C)	160A (2C)	240A (2C)	320A (2C)
Max. Continuous discharging current	240A (6C)	400A (5 C)	480A (4C)	480A (3C)	240A (6C)	400A (5C)	480A (4C)	480A (3C)
Working Temperature	Charging: 0°C~ 45°C; Discharging: -20°C~ 60°C							
Best Working Temperature	15°C~ 35°C							
Storage temperature	-40°C~ 60°C; (System SOC: Under 20%~40%; Within 1 month, -40°C~ 45°C; Within 6 month, -20°C~ 35°C)							
Working environment	Altitude: < 2500m; Relative humidity: < 95% (no condensation)							
System cooling mode	Controlled air cooling							
Insulation resistance	> 500MΩ@1500VDC							
Isolation and withstand voltage	2500VDC/1min							
Internal power supply mode	24VDC(Built-in DC/DC)							
Static power consumption	35W(Typical value, module fan not included)							
Data display	7" TN true color LCD screen (Ratio: 5:3, resolution ratio 800×480)							
IP rating	IP21							
Dimension								
W(mm)	600	600	600	600	1050	1050	1050	1050
D(mm)	550	750	950	1100	550	750	950	1100
H(mm)	2000	2000	2000	2000	1700	1700	1700	1700
Weight(Kg)	400	600	800	1000	630	950	1250	1600
Life cycle	3000 times (1C@25°C charging&discharging @100%DOD, EOL80%)							
Battery Management System (BMS)								
Cell voltage acquisition range	1-5V							
Cell voltage acquisition accuracy	≤ 10mV							
Total voltage acquisition accuracy	±1V or ±1%							
Current acquisition range	0~±250ADC	0~±500ADC	0~±625ADC	0~±750ADC	0~±250ADC	0~±500ADC	0~±625ADC	0~±750ADC
Current acquisition accuracy	≤±1°C							
Temperature acquisition range	-20 ~ 125°C							
Temperature acquisition accuracy	≤±1°C							
SOC estimation accuracy	≤ 8%							
SOH estimation accuracy	≤ 8%							
Charging over current protection	> 2C, 10s; > 2.2C, 5s; > 2.5C, 1s							
Discharging over current protection	> 6.2C,10s; > 6.5C,1s	> 5.2C,10s; > 5.5C,1s	> 4.2C,10s; > 4.5C,1s	> 3.2C,10s; > 3.5C,1s	> 5.2C,10s; > 5.5C,1s	> 5.2C,10s; > 5.5C,1s	> 4.2C,10s; > 4.5C,1s	> 3.2C,10s; > 3.5C,1s
Over temperature protection	Discharging > 60°C (20s) ; Charging > 45°C (20s)							
Low temperature protection	Discharging < -20°C (1s) ; Charging < 0°C (1s)							
Communication mode	CAN, RS485, dry contact							
Certification	IEC62619/IEC62620/EN61000/UN38.3							

High energy system



EBC480/50
Battery capacity **50Ah**
Rated voltage **480 Vdc**
Dimension **600*550*2000** (W*D*H)mm

EBC480/100
Battery capacity **100Ah**
Rated voltage **480 Vdc**
Dimension **600*750*2000** (W*D*H)mm

EBC480/150
Battery capacity **150Ah**
Rated voltage **480 Vdc**
Dimension **600*950*2000** (W*D*H)mm

EBC480/200
Battery capacity **200Ah**
Rated voltage **480 Vdc**
Dimension **600*1100*2000** (W*D*H)mm

EBC768/50
Battery capacity **50Ah**
Rated voltage **768 Vdc**
Dimension **1050*550*1700**(W*D*H)mm

EBC768/100
Battery capacity **100Ah**
Rated voltage **768 Vdc**
Dimension **1050*750*1700**(W*D*H) mm

EFS768-150
Battery capacity **150Ah**
Rated voltage **768 Vdc**
Dimension **1050*950*1700**(W*D*H)mm

EBC768/200
Battery capacity **200Ah**
Rated voltage **768 Vdc**
Dimension **1050*1100*1700**(W*D*H)mm



Model	EBC480/50	EBC480/100	EBC480/150	EBC480/200	EBC768/50	EBC768/100	EBC768/150	EBC768/200
Battery capacity	50Ah	100Ah	150Ah	200Ah	50Ah	100Ah	150Ah	200Ah
Rated voltage	480V	480V	480V	480V	768V	768V	768V	768V
Rated capacity	24kWh	48kWh	72kWh	96kWh	38.4kWh	76.8kWh	115.2kWh	153.6kWh
Battery voltage range(V)	420V～547.5V (Single Cell 2.8～3.65)				672V～876V (Single Cell 2.8～3.65)			
Max. Continuous charging current	50A (1C)	100A (1C)	150A (1C)	200A (1C)	50A (1C)	100A (1C)	150A (1C)	200A (1C)
Max. Continuous discharging current	100A (2C)	200A (2C)	300A (2C)	400A (2C)	100A (2C)	200A (2C)	300A (2C)	400A (2C)
Working Temperature	Charging: 0°C～45°C; Discharging: -20°C～60°C							
Best Working Temperature	15°C～35°C							
Storage temperature	-40°C～60°C; (System SOC: Under 20%~40%; Within 1 month, -40°C～45°C; Within 6 month, -20°C～35°C)							
Working environment	Altitude: ＜2500m; Relative humidity: ＜95% (no condensation)							
System cooling mode	Controlled air cooling							
Insulation resistance	＞500MΩ@1500VDC							
Isolation and withstand voltage	2500VDC/1min							
Internal power supply mode	24VDC(Built-in DC/DC)							
Static power consumption	35W(Typical value, module fan not started)							
Data display	7" TN true color LCD screen (Ratio: 5:3, resolution ratio 800×480)							
IP rating	IP21							
Dimension								
W(mm)	600	600	600	600	1050	1050	1050	1050
D(mm)	550	750	950	1100	550	750	950	1100
H(mm)	2000	2000	2000	2000	1700	1700	1700	1700
Weight(Kg)	450	650	850	1050	700	1000	1350	1650
Life cycle	4000 times (1C@25°C charging&discharging @100%DOD, EOL80%)							
Battery Management System (BMS)								
Cell voltage acquisition range	1-5V							
Cell voltage acquisition accuracy	≤10mV							
Total voltage acquisition accuracy	±1V or ±1%							
Current acquisition range	0~±250ADC	0~±500ADC	0~±625ADC	0~±750ADC	0~±250ADC	0~±500ADC	0~±625ADC	0~±750ADC
Current acquisition accuracy	≤±1°C							
Temperature acquisition range	-20～125°C							
Temperature acquisition accuracy	≤±1°C							
SOC estimation accuracy	≤8%							
SOH estimation accuracy	≤8%							
Charging over current protection	＞1C, 10s;							
Discharging over current protection	＞2.2C, 10s;							
Over temperature protection	Discharging＞60°C (20s); Charging＞45°C (20s)							
Low temperature protection	Discharging＜-20°C (1s); Charging＜0°C (1s)							
Communication mode	CAN, RS485, dry contact							
Certification	IEC62619/IEC62620/EN61000/UN38.3							

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